

Water hyacinth in. 200-bar vehicle-grade BioCNG out. Autonomously. At the bus terminal. No pipeline. No import. No chemist.

1. PROBLEM STATEMENT

Bangladesh's 500,000+ CNG vehicles run on imported fuel paid in scarce foreign exchange. Meanwhile, water hyacinth — an invasive weed with zero economic value — chokes 30%+ of the country's waterways. No local alternative fuel exists. No solution today connects these two crises.

2. TARGET USERS

- CNG auto-rickshaw & bus drivers — Dhaka, Chittagong, Sylhet, Barisal
- Urban transport terminal operators
- BRTC and private CNG fleet operators
- Global: Vietnam, Nigeria, Myanmar, Pakistan

3. PROPOSED SOLUTION

A shipping-container-sized Edge-AI-controlled BioCNG micro-refinery. Four integrated stages:

- Anaerobic digestion (water hyacinth feedstock)
- H2S scrubbing (iron sponge, >99% removal)
- CO2 membrane separation (>=95% CH4 purity)
- Multi-stage compression (200 bar output)

Deployed at bus terminals. AI replaces the operator.

4. KEY DIFFERENTIATORS

- Edge-AI: autonomous 24/7 pH/temp/gas control
- Zero supply chain: feedstock from adjacent canal
- Full end-to-end in one container — no off-site step
- Aquavoltaic FPV array: load-shedding proof
- 100% local build: Nawabpur + Dholaikhal, Dhaka

5. CURRENT DEVELOPMENT STATUS

- System architecture & AIoT control design: DONE
- Functional Edge-AI dashboard prototype: LIVE
merry-pudding-49b37d.netlify.app
- Component BOM (13 items): finalised
- Nawabpur procurement: in progress
- Dholaikhal fabrication order: placed
- Phase 1 CAPEX: BDT 1,46,500 (~\$1,330 USD)

6. EXPECTED IMPACT

- 20% cheaper fuel for 500,000+ drivers
- Eliminates forex drain on CNG imports
- Free waterway remediation (hyacinth cleared)
- CO2 offset: 15,000 tonnes/yr at 200-unit scale
- Energy access during grid load-shedding (24/7)
- Replicable: Myanmar, Vietnam, Nigeria, Pakistan

7. MARKET OPPORTUNITY

Bangladesh TAM: ~\$880M/yr (CNG vehicle fuel)
BioNode EaaS target: 5% share by Year 5 = \$44M ARR
Global TAM (invasive biomass markets): \$2.8B by 2032
Carbon credits: \$225K-\$375K/yr at Dhaka pilot scale
Model: own-operate-sell. Recurring daily revenue.
First 3 units: Gabtoli, Sayedabad, Mohakhali — Dhaka

8. FUTURE ROADMAP

Jul 2026 Prototype demo — BEAR Summit, Dhaka
Jan 2027 3-unit pilot live. Daily revenue. CES 2027
2027 20-unit Dhaka rollout. Series A raise
2028 Regional: Myanmar, Vietnam, Nigeria
2030 500 units. Carbon platform. Global OEM deals
2036 BioCNG infrastructure for 1 billion people

500K+	95%	200 bar	>95%	\$880M	Day 1
CNG vehicles in Bangladesh	Fuel import dependency	BioCNG output pressure	CH4 purity target	Bangladesh TAM	Revenue from deployment